

# Zhirui Dai

+1 (858) 200 5325 - [zh dai@ucsd.edu](mailto:zh dai@ucsd.edu) - [daizhirui.github.io](https://github.com/daizhirui) - [Google Scholar](#) - [LinkedIn](#) - [GitHub](#)

## RESEARCH INTERESTS

My research builds the spatial memory of embodied agents: real-time, scalable 3D scene representations that unify geometry (signed distance functions), occupancy, appearance and open-vocabulary semantics in a single adaptive-resolution octree, and that robots can query for safe motion planning, navigation and mobile manipulation. I am currently extending these representations to dynamic, changing environments. I have also worked on language-driven task planning with formal guarantees. A recurring theme of my work is making these methods fast enough to run onboard, through C++, custom CUDA kernels and open-source software.

## EDUCATION

### University of California San Diego

Ph.D. in Electrical and Computer Engineering; Advisor: Prof. Nikolay Atanasov

San Diego, CA, USA

01/2022 - Expected 06/2027

### University of California San Diego

M.S. in Electrical and Computer Engineering; Research with Prof. Nuno Vasconcelos

San Diego, CA, USA

09/2019 - 06/2021

### Fudan University

B.S. in Physics

Shanghai, China

09/2015 - 06/2019

## HONORS & AWARDS

|                                                                                                  |      |
|--------------------------------------------------------------------------------------------------|------|
| Best Paper Award, RSS 2025 Workshop on Leveraging Implicit Methods for Aerial Autonomy           | 2025 |
| Best Paper Nomination, RSS 2025 Workshops on Mobile Manipulation & Learned Robot Representations | 2025 |
| Electrical and Computer Engineering Department Fellowship, UCSD                                  | 2022 |
| Outstanding Graduate, Fudan University                                                           | 2019 |

## PUBLICATIONS

Underline: this author; \*: equal contribution. Google Scholar: [profile](#).

### Journal & Conference Papers

- Z. Dai\*, T. Fan\*, M. Amani, J. Seo, K. M. B. Lee, H. Oh, N. Atanasov. “Kernel-SDF: An Open-Source Library for Real-Time Signed Distance Function Estimation using Kernel Regression.” IEEE Robotics and Automation Letters (RA-L), 2026. [Paper Project](#)
- Z. Dai\*, Q. Qian\*, T. Fan, N. Atanasov. “OREN: Octree Residual Network for Real-Time Euclidean Signed Distance Mapping.” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. [Paper Project](#)
- T. Matthews, M. Kumar A. R., A. Begzadić, S. Tonkens, Z. Dai, N. Atanasov, J. Cortés, S. Herbert. “Expanding Safe Sets with Learning-Based Barrier Functions and Reachability.” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. [Paper](#)
- Z. Dai, H. Shin, Y. Tian, K. M. B. Lee, N. Atanasov. “Learning Scene-Level Signed Directional Distance Function with Ellipsoidal Priors and Neural Residuals.” IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2026. [Paper Project](#)
- S. Kim, W. Chung, Z. Dai, D. Bhatt, A. Shukla, H. Su, Y. Tian, N. Atanasov. “Seeing the Bigger Picture: 3D Latent Mapping for Mobile Manipulation Policy Learning.” IEEE International Conference on Robotics and Automation (ICRA), 2026. [Paper Project](#)
- B. Rabiei, M. Kumar A. R., Z. Dai, S. L. S. R. Pilla, Q. Dong, N. Atanasov. “LTLCodeGen: Code Generation of Syntactically Correct Temporal Logic for Robot Task Planning.” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025. [Paper Project](#)
- K. Long, Y. Yi, Z. Dai, S. Herbert, J. Cortés, N. Atanasov. “Sensor-Based Distributionally Robust Control for Safe Robot Navigation in Dynamic Environments.” The International Journal of Robotics Research (IJRR), 2025. [Paper Project](#)
- Z. Dai, A. Asgharivaskasi, T. Duong, S. Lin, M.-E. Tzes, G. J. Pappas, N. Atanasov. “Optimal Scene Graph Planning with Large Language Model Guidance.” IEEE International Conference on Robotics and Automation (ICRA), 2024. [Paper Project](#)
- Z. Dai, Y. Jiang, Y. Li, B. Liu, A. B. Chan, N. Vasconcelos. “BEV-Net: Assessing Social Distancing Compliance by Joint People Localization and Geometric Reasoning.” IEEE/CVF International Conference on Computer Vision (ICCV), 2021. [Paper](#)

### Preprints

- Z. Dai, Q. Qian, D. M. Nguyen, Q.-D. Pham, K. Bronder, C. Nieto-Granda, Y. Chen, Q. Nguyen, N. Atanasov. “OREN-X: Octree Residual Network for Real-Time Multi-Modal Mapping.” Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027, under review. [Paper Project](#)
- Q.-D. Pham, A. Dao, D. V. Le, N. V. T. Pham, T. A. Nguyen, Z. Dai, Y. Chen, T. P. Le, V.-T. Nguyen, Q. Nguyen. “AdaGeoVLN: Selective Geometry Across Representation Depth and Navigation Time for Vision-Language Navigation.” Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027, under review. [Paper](#)
- J. Stanley\*, Z. Dai\*, Q. Qian, T.-C. Ho, T. Fan, S. Saha, C. Barnrover, K. M. B. Lee, N. Atanasov. “From Distances to Trajectories: Real-Time Signed Distance Function Mapping and Distance-Accelerated Motion Planning for UAVs.” Submitted to Autonomous Robots, 2026, under review. [Paper Project](#)
- K. M. B. Lee, Z. Dai, C. Le Gentil, L. Wu, N. Atanasov, T. Vidal-Calleja. “Safe Bubble Cover for Motion Planning on Distance Fields.” arXiv preprint arXiv:2408.13377, 2024. [Paper](#)

### Workshop Papers

- J. Stanley\*, Z. Dai\*, T.-C. Ho, S. Saha, C. Barnrover, N. Atanasov. “From Distances to Trajectories: Real-Time Signed Distance Function Mapping and Distance-Accelerated Motion Planning for UAVs.” ICRA 2026 Workshop on Aerial Inspection for Marine Infrastructures (AIMI), 2026. [Paper Project](#)

- [Z. Dai](#), H. Shin, Y. Tian, K. M. B. Lee, N. Atanasov. “**Learning Scene-Level Signed Directional Distance Function for Aerial Autonomy.**” Workshop on Leveraging Implicit Methods for Aerial Autonomy at RSS, 2025. [Paper](#) [Project](#) [Best Paper Award]
- S. Kim, W. Chung, Y. Tian, [Z. Dai](#), A. Shukla, H. Su, N. Atanasov. “**Seeing the Bigger Picture: 3D Latent Mapping for Mobile Manipulation Policy Learning.**” Workshop on Mobile Manipulation and Workshop on Learned Robot Representations at RSS, 2025. [Paper](#) [Project](#) [Best Paper Nomination]

## RESEARCH EXPERIENCE

### Research Intern, VinMotion Inc.

Irvine, CA, USA

Real-Time Multi-Modal 3D Mapping (in collaboration with ERL, UCSD)

06/2026 - 09/2026

#### • [OREN-X](#): Real-Time Multi-Modal Mapping

Submitted to ICRA 2027

- Developed a unified octree representation that jointly learns SDF, occupancy, radiance and vision-language fields online, with cross-modality supervision and online dictionary learning (3.7x feature compression); 80+ fps for SDF and 30+ fps for all modalities.
- Improved near-surface SDF accuracy by 33% (over OREN) and open-vocabulary 3D mIoU by 71%; validated on Replica, TUM RGB-D, a Jackal robot and a humanoid across different scene scales (studio, lab space and construction site).

### Graduate Student Researcher, Existential Robotics Lab, UCSD

San Diego, CA, USA

Advisor: Prof. Nikolay Atanasov

01/2022 - Present

#### • Distance-Field Representations for Mapping & Planning

TPAMI 2026, IROS 2026, RA-L 2026

- Introduced neural signed directional distance functions ([SDDF](#)) with ellipsoidal priors and neural residuals, enabling single-pass occlusion reasoning and collision checking with SOTA accuracy on Replica.
- Developed [OREN](#), an octree residual network for globally accurate real-time SDF learning from LiDAR/RGB-D (7 fps on Jetson Orin).
- Developed [Kernel-SDF](#) for SDF estimation with uncertainty quantification, and distance-accelerated UAV motion planning ([Bubble\\*](#)) deployed onboard a quadrotor.

#### • 3D Vision-Language Maps for Mobile Manipulation

ICRA 2026

- Co-developed [SBP](#), a 3D latent map of VLM features that serves as spatiotemporal memory for a transformer manipulation policy, improving success by up to 15% over image-based policies.

#### • Language-Driven Task Planning with Guarantees

ICRA 2024, IROS 2025

- Combined LLM guidance with consistent LTL heuristics for [optimal scene graph planning](#) (1600x speedup over AMRA\*), and developed [LTLCodeGen](#) for syntactically correct LTL generation.

### Graduate Student Researcher, Statistical Visual Computing Lab, UCSD

San Diego, CA, USA

Advisor: Prof. Nuno Vasconcelos

06/2020 - 08/2021

#### • Geometric Reasoning for Social Distance Monitoring

ICCV 2021

- Proposed [BEV-Net](#), jointly localizing people and reasoning about geometry via a bird's-eye-view projection layer with calibration-free camera pose recovery.

## TEACHING & MENTORING

Teaching Assistant, [ECE276B: Planning & Learning in Robotics](#), UCSD

Spring 2023, Spring 2024

Research mentor, Qihao Qian (M.S.): co-first author of OREN (IROS 2026); co-author of OREN-X

02/2025 - Present

Research mentor, Tianxing Fan (M.S.): co-first author of Kernel-SDF (RA-L 2026); co-author of OREN

12/2024 - 02/2026

Mentor, [ENLACE](#), [SRIP](#) and [STARS](#) Summer Research Programs, UCSD

07/2022 - 08/2022

Mentor, [Guided Engineering Apprenticeship in Research \(GEAR\)](#), UCSD

09/2021 - 07/2022

## PROFESSIONAL SERVICE

**Conference Reviewer:** NeurIPS, CVPR, RSS, CoRL, ICRA, IROS

**Journal Reviewer:** IEEE Robotics and Automation Letters (RA-L), IEEE Trans. on Intelligent Transportation Systems (T-ITS)

## OPEN-SOURCE SOFTWARE

**Released:** [erl\\_geometry](#) (C++/CUDA semi-sparse octree), [Kernel-SDF](#) (C++/Python/ROS), [neural\\_sddf](#) (PyTorch + CUDA ops), [BEV-Net](#)

**Contributions to:** [PyTorch](#), [Open3D](#), [PointCloudLibrary](#), [pybind11](#)

**Skills:** C++, CUDA, Python, PyTorch, ROS, OpenCV, Open3D, Docker, Kubernetes, ManiSkill

## INDUSTRY EXPERIENCE

Research Intern, XCOM Labs Inc. (neural pose prediction for VR/AR remote rendering)

08/2021 - 12/2021

Software Development Intern, Camel Microelectronics Inc. (SoC IDE & embedded toolchain)

01/2018 - 06/2019